Two-Stage Registration of SAR Images With Large Distortion Based on Superpixel Segmentation

Deliang Xiang, *Member, IEEE*, Xiaoyu Pan, Huaiyue Ding, Jianda Cheng, *Member, IEEE*, and Xiaokun Sun

***Abstract*— When the geometric distortion of the synthetic aperture radar (SAR) images to be registered is large, the spatial correspondence between the feature points of the two images will change significantly. Hence, the registration of SAR images with large geometric distortion is challenging. To solve this problem, a two-stage registration method of SAR images with large distortion based on superpixel segmentation is proposed in this article. First, the two SAR images are coarsely registered by geographic coordinate referencing. After coarse registration, superpixel segmentation is performed on the two SAR images, respectively. Next, in the superpixel neighborhood of the reference image, we slide the corresponding superpixel template of the sensed image, finding its position with the highest similarity in the reference image. Compared with the traditional fixed-size template, the superpixel template can segment the distorted region more effectively. Meanwhile, with the help of the adaptive threshold detector proposed in this article, the regions with varying degrees of distortion can be distinguished based on their similarity. Furthermore, the geometry mapping relationship is calculated for the regions with different distortion degrees in the images, respectively, and the corresponding feature points in different images are accurately matched to complete the fine registration. Finally, the registration results of regions with different degrees of distortion are fused to obtain the final SAR image registration results. Experimental results based on Sentinel-1 data show that the registration accuracy of the proposed method can reach within 1 pixel.**

***Index Terms*— Geometric distortion, superpixel segmentation, synthetic aperture radar (SAR) images, two-stage registration.**

I. Introduction

SYNTHETIC aperture radar (SAR) has advantages of all-day, all-weather imaging capability and high spatial resolution, which is widely used in geological exploration, environmental monitoring, land use, urban planning, water resource management, forest management, disaster monitoring, and emergency response, among other fields [1], [2], [3], [4]. The registration of SAR images is a crucial step in radar remote sensing image interpretation. It involves correlating feature points between different SAR images to achieve spatial geometric transformations. However, imaging at different times, with different sensors, and from different angles can result in large geometric distortions in SAR images being registered. In addition, the inherent speckle noise in SAR images increases the difficulty of registration. Therefore, SAR image registration remains a current research focus in radar remote sensing image interpretation [5].

Currently, image registration methods are broadly classified into two categories, area-based methods and feature-based methods [6]. Area-based methods estimate the transformation relationship between two images by utilizing similarity measures and using template matching [7]. Commonly used similarity metric functions include mutual information (MI) [8], sum of squared differences (SSDs) [9], and normalized cross correlation (NCC) [10]. Area-based registration methods are relatively simple and do not require complex preprocessing of the sensed image and reference image. They only rely on gray-scale information to complete the registration. However, this method makes it difficult to correct the nonlinear transformations between the images. Furthermore, it is sensitive to geometric and radiometric differences as well as speckle noise in SAR images, leading to poor registration results.

Compared with area-based registration methods, feature-based methods have lower computational complexity and greater robustness. The feature-based registration methods are mainly divided into extracting features, constructing feature descriptors, feature matching, and calculating a transformation model, thus achieving high-precision image registration [11], [12], [13], [14]. Among numerous features, point features are most widely used, and the most representative registration algorithm is the scale-invariant feature transform (SIFT) [15]. Due to its scale and rotation invariance [16], [17], it is widely used in optical remote sensing image registration [18]. The speckle noise in SAR images causes the feature points detected by the traditional SIFT algorithm to be highly unstable, resulting in numerous incorrect matches. Therefore, it is difficult to apply the SIFT method directly to SAR image registration.

In response to the above problems, Schwind et al. [19] proposed the SIFT-OCT algorithm, which achieves image filtering by omitting the first layer of the scale space pyramid in the SIFT algorithm and employing infinite impulse response filtering. This approach can maintain the correct number of matching points while enhancing algorithm efficiency. Fan et al. [20] combined the SIFT algorithm with sparse representation techniques, utilizing a spatial correlation strategy based on stationary wavelet transform to extract feature points from the SIFT algorithm. This approach reduces the impact of speckle interference in SAR images. Dellinger et al. [21] proposed the SAR-SIFT algorithm with a new definition of the

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gradient specifically for SAR registration. Wang et al. [22] proposed the bilateral filter scale-invariant feature transform (BFSIFT) algorithm to improve issues like Gaussian blurring and indistinct image details caused by Gaussian filtering in Gaussian scale space (GSS). Wang et al. [23] solved the problem of edge detail blurring caused by anisotropic Gaussian filters using the adaptive anisotropic Gaussian SIFT (AAG-SIFT) algorithm, enhancing SAR image registration accuracy and efficiency. Ma et al. [24] proposed to use of the SAR-SIFT algorithm to extract feature points from SAR coherent intensity maps and match them according to their spatial constraints. Although the above methods can achieve SAR image registration, they still fail to detect effective keypoints in complex SAR imaging scenes. For example, SAR images in mountainous terrains may present geometric distortions like perspective contraction, overlapping, and others. Moreover, when the imaging angles of sensors are inconsistent or when imaging platforms vary in height, SAR images of the same area may exhibit significant geometric differences. These factors can result in erroneous matching of keypoints between the reference and sensed images, making accurate registration challenging.

In addition, the above SIFT-based point feature extraction methods involve manually designed features. The feature detection and feature description of them are relatively sparse, which can only extract low-level semantic information [25]. In SAR image feature extraction with large geometric distortion, it is difficult to obtain accurate feature points. Paul and Pati [26] designed a weight function to select SAR-SIFT features with higher saliency. Simultaneously, they constructed multiple concentric support regions at different scales to build more efficient and robust feature descriptors. With the development of deep learning, researchers have proposed some convolutional neural networks (CNN)-based deep feature extraction methods, such as Siamese networks [27], pseudo-Siamese networks [28], and so on, to extract richer features [29], [30]. These CNN-based methods can adapt to the geometric and radiometric differences between SAR images while mining nonlinear relationships among them. However, as the convolutional layers deepen, details in SAR images tend to be gradually lost, and the preserved prominent features often appear in regions with higher backscattering intensity, such as man-made structures and forested areas, where large geometric distortions may exist. In SAR images, nonlinear radiometric distortions and speckle noise can affect the accuracy of feature point localization, deep neural networks still struggle to effectively eliminate interference points. In addition, deep learning-based SAR image registration requires a large set of well-registered paired datasets for training, which is difficult to acquire in practical applications, leading to high costs associated with manual annotation.

The superpixel segmentation method was initially proposed by Ren and Malik [31] for optical color images. Li et al. [32] introduced a graph-based segmentation method using normalized cuts. Subsequently, Achanta et al. [33] proposed a k-means clustering algorithm called simple linear iterative clustering (SLIC) for superpixel segmentation. The SLIC algorithm utilizes feature vectors composed of color values and pixel positions. Based on this, an increasing number of methods based on the SLIC framework have been developed. Akyilmaz and Leloglu [34] achieved good results by replacing Euclidean distance with ratio similarity. The method proposed by Ban et al. [35] associates superpixels with Gaussian distributions, where each pixel is represented by a random variable and described by a weighted sum of several Gaussian functions, thereby segmenting superpixels of similar sizes.

Due to the significant influence of speckle noise on SAR images, directly applying the above algorithms to generate superpixels for SAR images may not yield satisfactory results. Therefore, some studies have optimized and improved methods specifically for the characteristics and problems of SAR images. Liu et al. [36] and Qin et al. [37] proposed a superpixel segmentation method for polarimetric SAR images, improving the SLIC algorithm by enhancing cluster center initialization and postprocessing steps, and using the Wishart distance instead of the Euclidean distance. Xiang et al. [38] proposed an improved version of SLIC superpixels for SAR images, utilizing a distance metric based on intensity and positional similarity [39]. The method proposed by Arisoy and Kayabol [40] utilizes the amplitude and pixel coordinates of SAR images as features. It statistically models the feature vectors using the statistical properties of SAR images and employs a finite mixture model to cluster pixels into superpixels.

Aiming at the above problems, in this article, a new method for large distortion SAR image registration based on superpixel segmentation is proposed. First, the reference image and the sensed image are coarsely registered by geographic coordinate reference. Building upon this, superpixel segmentation is performed on them to obtain corresponding superpixels. Next, we use the superpixels from the sensed image as templates, allowing them to traverse the neighborhood of the corresponding superpixels in the reference image. Combining the adaptive threshold detector designed in this article, we distinguish regions with different degrees of distortion. For them, taking the geometric center of superpixels as feature points, transformation parameters are calculated separately, and fuse the registration results from different regions to obtain the final finely registered result. The main contributions of this article are as follows.

1) This article proposes a superpixel-based SAR image registration method, which involves segmenting the reference and sensed images into superpixels, thereby completing the selection and fine registration of different degrees of distortion regions in SAR images. The traditional fixed-size template may lead to matching errors due to the inconsistency in the distortion degrees of internal pixels. In contrast, superpixels can effectively segment distorted regions, resulting in higher registration accuracy.
2) A method to measure the similarity of superpixel templates is proposed. With the transformation relationship obtained from the coarse registration, the superpixel labels of the reference image are mapped to the corresponding positions of the sensed image. Then, the corresponding superpixels are found and the similarity judgment is performed.

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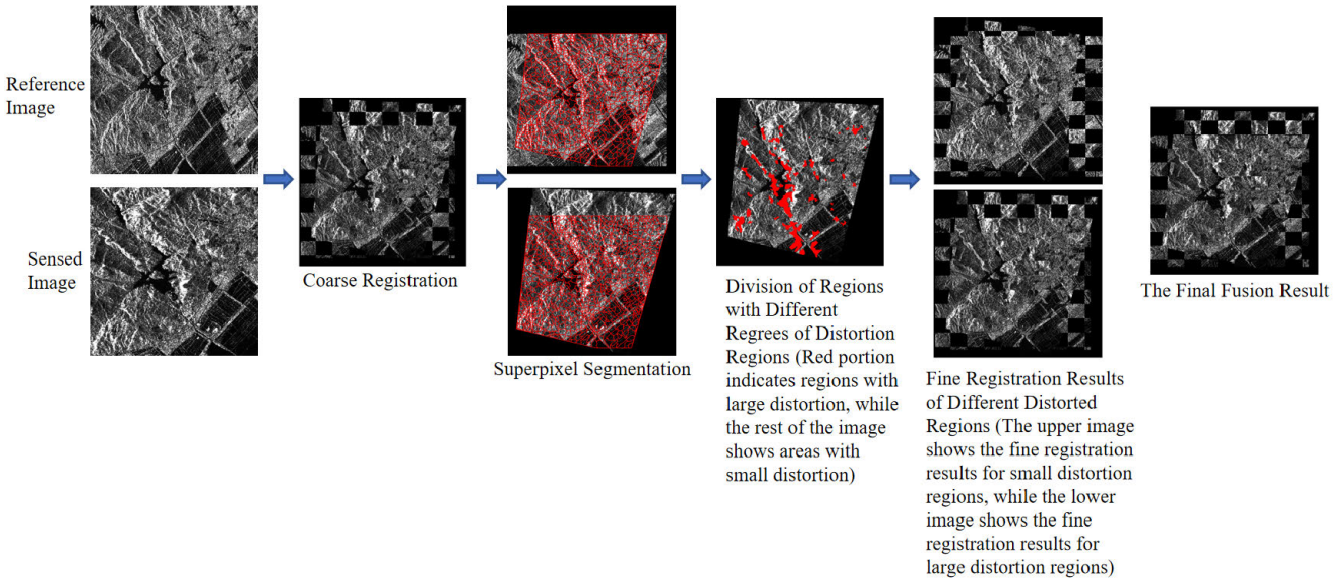


Fig. 1. Overall flowchart of the method in this article.

3) We design an adaptive threshold detector. It can judge the distortion degree of SAR images, allowing for registration according to varying degrees of distortion. It can set different distortion degree thresholds for different images. Thus, we can accurately identify large distortion regions in each image and extract more reliable feature points. Finally, registration accuracy can be improved.

The rest of the article is organized as follows. Section II describes the proposed method in detail. Section III gives the experimental results of our proposed method on SAR images of different sites and a comparison with other commonly used methods. Section IV shows the importance of two-stage registration. Finally, conclusions are presented in Section V.

II. Proposed Method

The main steps of the proposed method for registering SAR images with large distortion in this article include coarse registration using geographical coordinate references, superpixel segmentation, division of regions with different degrees of distortion, fine registration, and fusion, as illustrated in Fig. 1.

A. Coarse Registration by Geographical Coordinate Referencing

Coarse registration is a preprocessing step in this method, aiming to eliminate the rotation, translation, and scaling differences between SAR images, obtaining initial registration results. The coarse registration establishes an initial correspondence between the two images, reducing the search range of template matching during fine registration. This article utilizes the SAR image's inherent geographical coordinate references for coarse registration. We select four control points on the reference image and the sensed image, respectively. Two control points are chosen from regions with small geometric distortion, such as farmlands and bare soil areas, while the other two are chosen from regions with large geometric distortion, such as mountainous areas and urban areas. The projection transformation matrix is calculated based on the control points' coordinates, and the transformed sensed image is obtained for the subsequent input in fine registration.

B. Superpixel Segmentation

Superpixel aggregates pixels with similar gray-scale features together to process images, replacing individual pixels. This approach not only enhances image processing speed but also holds advantages in feature extraction. Superpixel has been widely utilized in the field of computer vision [41]. Traditional superpixel segmentation algorithms are designed for natural images, which are typically not applicable to SAR images due to the significant influence of speckle noise on superpixel generation results. After coarse registration on the reference and sensed images, we perform superpixel segmentation on both images in this article. For superpixel segmentation in SAR images, an improved edge-aware superpixel generation with one iteration merging (ESOM) [42], [43], [44] method which we previously proposed is employed. It utilizes a ratio-based edge detector to extract edge features from both the reference and sensed image, computing their edge strength values, which have been proven to be robust to multiplicative noise [39]. The detector window used in this article is Gaussian-Gamma double window [45], which shows higher adaptability and robustness compared to the Gaussian-shaped (GS) window used in the original method. Initially, SAR images are performed using Gaussian-Gamma double windows, effectively suppressing speckle noise interference. The formula for each window is as follows:

$$W(x,y)=\frac{|y|^{\alpha-1}}{\sqrt{2\pi}\sigma_x\Gamma(\alpha)\beta^{\alpha}}\cdot\exp\left(-\left(\frac{x^2}{2\sigma_x^2}+\frac{|y|}{\beta}\right)\right) \quad (1)$$

where σ_x is the Gaussian scale factor along the edge direction, it is mainly used to regulate the length of the window. α and β are the parameters of the perpendicular to the edge direction of the Gamma function. α is used to adjust the double-window spacing, and β is used to control the width of the window, so as

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to make the local focusing of the Gaussian-Gamma double window better. It is helpful for the extraction of nonideal edges of the algorithm.

Then, we convolve the window function with pixel intensity to obtain the local mean Ratio(x,y)

$$\text{Ratio}(x, y) = \min_{f=0,1,\ldots,N_f-1}\left\{\frac{m_1(x, y \mid \theta_f)}{m_2(x, y \mid \theta_f)}, \frac{m_2(x, y \mid \theta_f)}{m_1(x, y \mid \theta_f)}\right\} \quad (2)$$

where $m_1(x, y \mid \theta_f)$ is the local mean value information obtained by convolving the upper half-window in the direction θ_f, and $m_2(x, y \mid \theta_f)$ is the local mean value information obtained by convolving the lower half-window in the direction θ_f.

Based on this, the edge strength map (ESM) and edge directional map (EDM) can be computed, as shown in the following formulas:

$$\text{ESM} = 1 - \min_{\theta_f}\text{Ratio}(x, y) \quad (3)$$

$$\text{EDM} = \underset{\theta_f}{\text{argmin}}\,\text{Ratio}(x, y) + \pi/2. \quad (4)$$

The next step involves defining an edge-aware dissimilarity, which utilizes edge information to effectively attach superpixels to image boundaries, aiding in their generation. As pixels along edges acquire a higher edge-aware dissimilarity, they are less likely to merge into superpixels, resulting in better boundary adherence of the generated superpixels.

The method here employs the generalized likelihood ratio (GLR) as the intensity dissimilarity first, defined as

$$S_{\text{intensity}}(i, j) = 2M \cdot \ln\frac{I(i) + I(j)}{2\sqrt{I(i) \cdot I(j)}} \quad (5)$$

where i stands for the superpixel center and j denotes the pixel to be labeled. $I(i)$ represents the average intensity of i and its eight neighboring pixels, while M denotes the number of pixels in $I(i)$, with a value of 9.

The spatial dissimilarity is defined as

$$S_{\text{spatial}}(i, j) = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}. \quad (6)$$

Next, by combining intensity dissimilarity with spatial dissimilarity and incorporating edge information, the edge-aware dissimilarity between i and j is defined as

$$S(i, j) = S_{\text{intensity}}(i, j) + \lambda \cdot S_{\text{spatial}}(i, j) + \beta \cdot \text{EDGE}(x_j, y_j) \quad (7)$$

where the balancing factor λ is used to maintain a balance between spatial dissimilarity and intensity dissimilarity, $\beta \cdot \text{EDGE}(x_j, y_j)$ represents the edge-aware term, EDGE denotes the final edge map, and β stands for the weighting factor.

Finally, this method achieves natural connectivity among superpixels by leveraging the continuity of neighboring pixels and edge information. It simultaneously performs label assignment and updates the superpixel center within a single iteration merging process. Following the initial superpixel segmentation, adjacent superpixels can be merged based on their similarity, further enhancing the quality of the segmentation results.

While there are many superpixel segmentation methods, there are not many registration methods based on superpixels. In our proposed method, we first apply the aforementioned algorithm for superpixel segmentation to the reference image. Then, based on the coarse registration transformation, we map the superpixel segmentation result of the reference image onto the transformed sensed image, locating corresponding superpixels. Finally, we obtain the superpixel segmentation result of the sensed image. The superpixel segmentation results of the reference and sensed images are shown in Fig. 2.

C. Division of Regions With Different Degrees of Distortion

In order to achieve fine registration between SAR images, traditional template matching methods overlook the influence of regions with different degrees of distortion on the selection of feature points in the image, leading to a high rate of mismatching. This article proposes a template matching method based on superpixels and an adaptive threshold detector, which can more effectively segment distorted regions compared to a fixed-size window template. By computing the gray-scale similarity between corresponding superpixels in the reference and sensed images, each image is divided into two categories, regions with large and small geometric distortion degrees. Specific transformation matrices are then calculated for these two types of distortion regions. This article proposes to use the NCC as a similarity function, which compares the gray-scale information between the superpixels of the two SAR images to determine their similarity. The calculation formula is

$$\text{NCC}(I_1, I_2) = \frac{\sum_{i=1}^{n}(I_1(i) - \mu_1)(I_2(i) - \mu_2)}{\sqrt{\sum_{i=1}^{n}(I_1(i) - \mu_1)^2 \sum_{i=1}^{n}(I_2(i) - \mu_2)^2}} \quad (8)$$

where I_1 and I_2 are the reference image and the sensed image, respectively, n is the size of the superpixel template, which is the number of pixels within the template, and μ is the mean of pixels within the superpixel template.

First, after coarse registration, corresponding superpixels are located on the reference and sensed image. Using the superpixels from the sensed image as templates, a similarity calculation is performed by sliding the template within a 10-pixel neighborhood of the corresponding superpixels in the reference image, i.e., within a range of 10 pixels in each direction: up, down, left, and right. This process aims to find the best match, as illustrated in Fig. 3. For each pair of superpixels, the position with the highest NCC value is identified, representing the highest similarity between the superpixel in the reference image and its corresponding superpixel in the sensed image. A triplet movement is defined to record the template's translation direction, size, and maximum NCC value, as shown in the following equation:

$$\text{movement} = (p_m, q_m, l_m) \quad (9)$$

where p represents the movement in the vertical (up and down) direction, while q denotes the movement in the horizontal (left and right) direction, the absolute values of them indicate the distance of movement. When $p_m > 0$ is present, it indicates the template moving upward; when $p_m < 0$ is present, it signifies the template moving downward. $q_m < 0$ signifies

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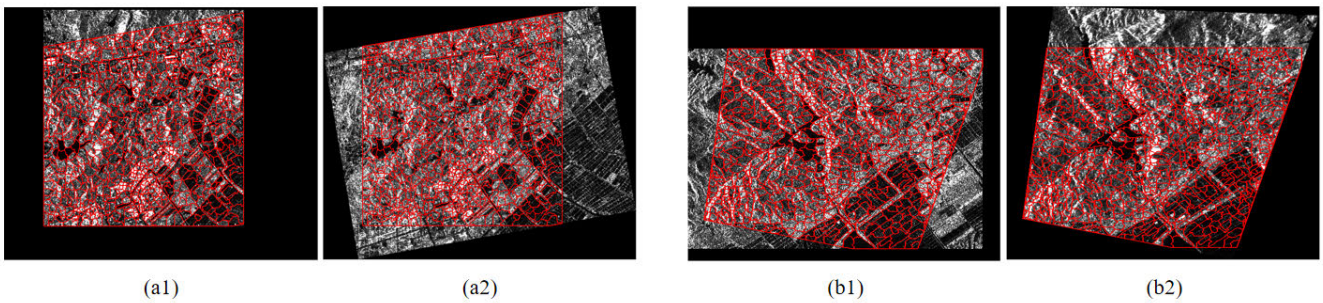


Fig. 2. Examples of superpixel segmentation result. (a1) and (b1) Reference images for two sites. (a2) and (b2) Sensed images for two sites.

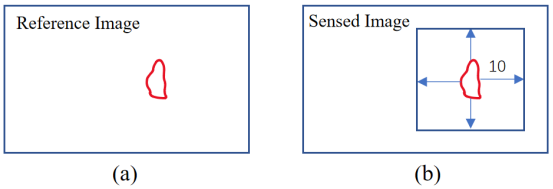


Fig. 3. Gray-scale similarity registration diagram. (a) Example of superpixel in the reference image. (b) Example of superpixel search in the sensed image.

the template moving left; $q_m > 0$ signifies the template moving right. l_m represents the maximum NCC value for each pair of superpixels, $m \in (1, 2, \ldots, M)$, and M represents the number of superpixels.

Subsequently, we traverse the entire image to obtain the similarity sequence, as shown in the following equation:

$$\text{List} = (l_1, l_2, \ldots, l_m). \tag{10}$$

Based on this, the article proposes an adaptive threshold detector to classify distortion regions in SAR images, as illustrated in Fig. 4. The threshold detector utilizes a sliding window approach for l_m value detection, distinguishing regions with different degrees of distortion based on l_m. When the l_m is lower than the threshold K, indicating a lower gray-scale similarity, the superpixel is identified as belonging to a region with larger geometric distortion. Conversely, when the l_m is higher than K, indicating higher gray-scale similarity, the superpixel is classified as a region with smaller geometric distortion, as depicted in Fig. 5. The threshold is calculated based on the average and standard deviation of the similarity of all corresponding superpixels between the SAR image pairs, adaptively reflecting the overall distortion degree of different images. The calculation formula is as follows:

$$K = \mu_{\text{List}} - 2\sigma_{\text{List}} \tag{11}$$

where μ_{List} is the mean of the sequence List, and σ_{List} is the standard deviation of the sequence List.

From Fig. 5, it can be observed that the adaptive threshold detector proposed in this article effectively divides SAR images into two categories, regions with large and small geometric distortion degrees. The red regions indicating large distortion are primarily distributed in mountainous regions and urban areas, attributed to the side-view imaging of the radar. Terrain variations cause image distortion, thus resulting in geometric distortion [46]. Conversely, flat farmlands and bare soil areas are categorized as small distortion regions. Furthermore, different imaging angles also contribute to geometric distortion. Therefore, a few flat areas marked in red might result from substantial differences in imaging angles between the reference and sensed images.

D. Fine Registration

To further enhance the accuracy of coarse registration, this article introduces an improved RANSAC algorithm. By removing mismatched points between the reference and sensed images, it obtains accurate matching keypoints, thereby achieving fine registration.

First, we map the keypoints from the reference image to the sensed image. The geometric centroids of superpixels from two types of distorted regions are selected as keypoints. Then, we utilize these keypoints to compute parameters for an affine transformation model and obtain their mapped points on the sensed image through the parameter matrix. Subsequently, we calculate the average error between these mapped points and the initial keypoints on the sensed image. Finally, we iterate this calculation until the average error between them is less than a threshold, thereby the mismatched points are removed. The formula for calculating the average error of keypoints is as follows:

$$\text{meanerr} = \frac{\text{norm}\left(\sqrt{(\triangle x)^2 + (\triangle y)^2}\right)}{\sqrt{\text{num}}} \tag{12}$$

where num is the number of input feature points, $\triangle x$ and $\triangle y$ are the x and y directional offset between the initial feature points and the mapped points on the sensed image, respectively, and norm denotes the calculation of the $L2$ norm.

After the initial keypoints undergo selection, they become keypoints. Using the keypoints from the two types of distorted regions, transformation matrices T_S and T_L are obtained through affine transformations. In this way, we further obtain the transformed sensed images of the two distorted regions as well as the registration result images.

E. Superpixel Replacement

Due to the computation of different transformation parameters from different sets of keypoints, registration results for two categories of distorted regions can be obtained. Furthermore, superpixel replacement yields the finely registered result for the entire SAR image. The transformed sensed image obtained from coarse registration is denoted as Result_0, the one from small distortion region registration is denoted as Result_S, and

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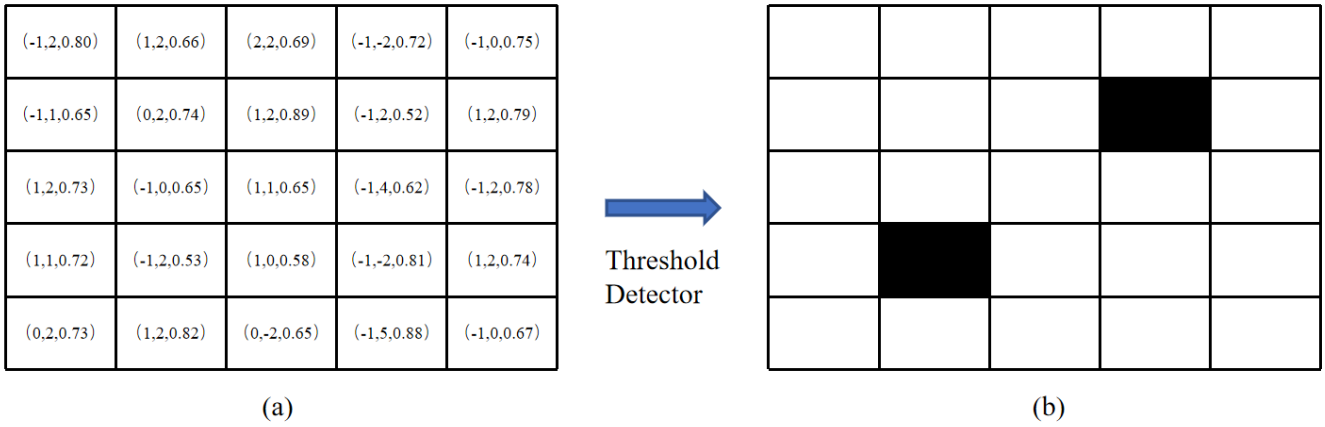


Fig. 4. Distortion region classification diagram. (a) Triplet is movement. (b) Example of distortion region classification results, where the black portion represents large distortion regions, and the white portion represents small distortion regions.

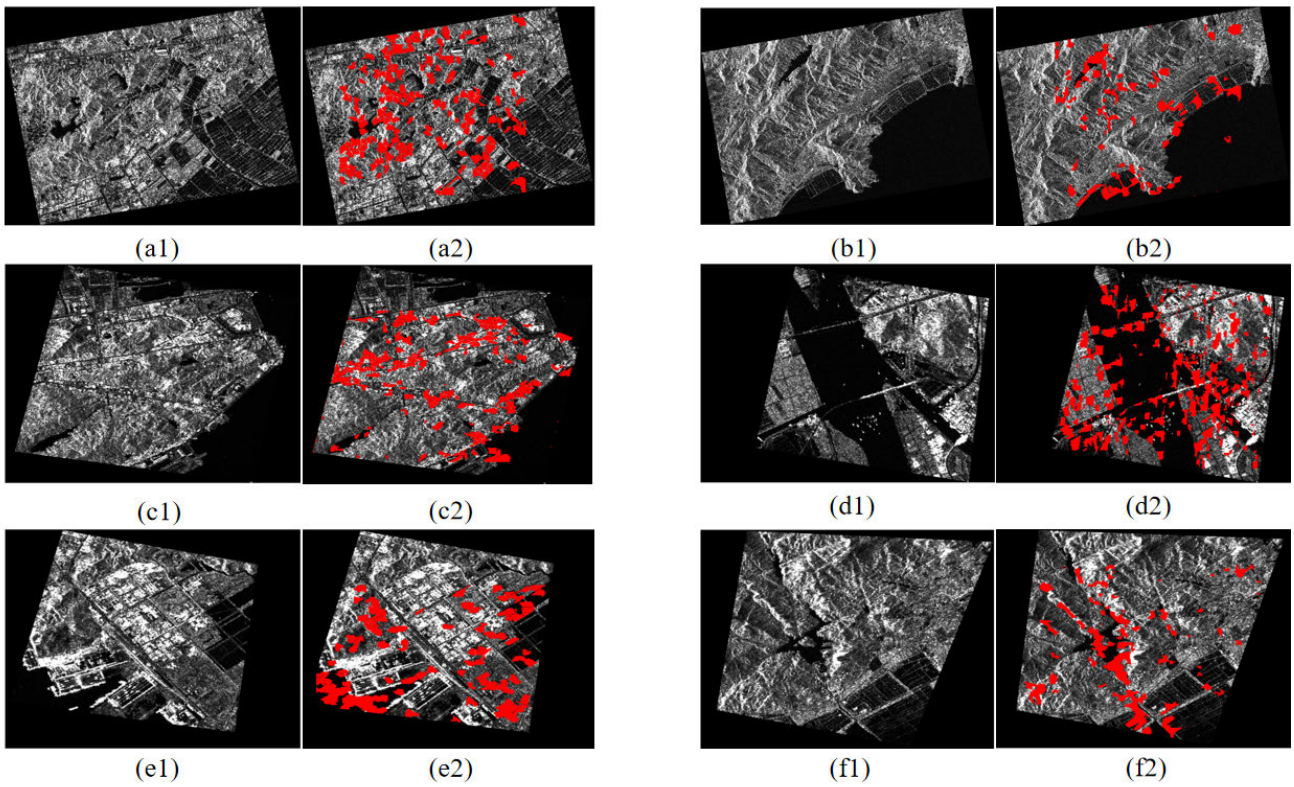


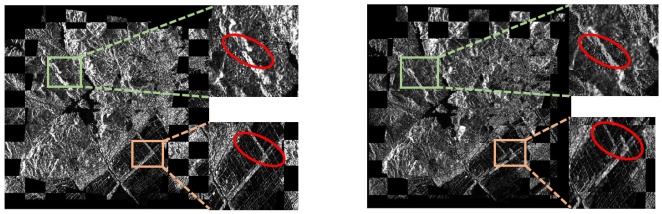
Fig. 5. Division results of different distortion degrees in the sensed images. (a1)–(f1) Sensed images of six sites. (a2)–(f2) Corresponding division results, where large distortion regions are depicted in red and the rest of the images indicate regions with small distortion.

the one from large distortion region registration is denoted as Result$_L$. As shown in Fig. 6, Result$_S$ demonstrates better registration in small distortion regions, while Result$_L$ shows superior registration in large distortion regions. Consequently, based on the transformed sensed images of these two region types and their superpixel segmentation results, the final transformed sensed image is achieved through superpixel replacement, as illustrated in Fig. 7.

Upon zoomed-in view on the selected region from Fig. 6(a), it can be observed that the content edges of farmland with small distortion are well connected. However, in the mountainous regions with large distortion, there are discontinuities. This indicates that the registration result for small distortion regions in Result$_S$ is better, while it's poorer in the large distortion regions. Conversely, Fig. 6(b) demonstrates the opposite, where Result$_L$ shows better registration for large distortion regions and poorer registration for small distortion regions. Therefore, segmenting and merging SAR images based on distortion degrees can achieve more fine registration, especially for large distortion regions.

Based on transformation matrices T_S and T_L in this article, we first obtain Result$_S$ and Result$_L$ from Result$_0$, respectively. Next, using the method of superpixel segmentation described in Section II-B, we calculate the superpixel segmentation

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(a) (b)

Fig. 6. Comparison example of the registration checkboard overlay for two types of distorted regions. The green box represents large distortion regions, while the orange box signifies small distortion regions. (a) Overlay result of the reference image with Result$_S$. (b) Overlay result of the reference image with Result$_L$.

results for Result$_S$ and Result$_L$, respectively. Then, we retain the superpixels corresponding to small distortion regions in Result$_S$ and replace the superpixels in Result$_S$ that are marked in red denoting large distortion regions, with the corresponding superpixels from Result$_L$. Ultimately, the registration result by merging these two types of regions is obtained, and we achieve fine registration for large distortion SAR images.

III. Experiments and Analysis

A. Data Description and Parameter Settings

This article conducts algorithm performance testing using spaceborne Sentinel-1 SAR images. The SAR data is acquired from Sentinel-1's C-band data over the Greater Bay Area of China (22°03′44″ N, 113°24′47″ E) with a resolution of 14 × 14 m. In addition, we perform basic orbit correction, radiometric calibration, and terrain correction on the SAR images. Manual registration of SAR images is carried out using ENVI software. As depicted in Fig. 8, this article selects SAR images from six different sites and Table I provides detailed information regarding the experimental data.

From the six pairs of SAR images, it can be seen that Fig. 8(a) and (c) have a rich variety of land cover types, including farmland and mountainous areas. As shown in Fig. 9(b), due to radar side-view imaging, mountainous areas exhibit significant geometric variations. Fig. 8(b), (d), and (e) exhibits significant changes in acquisition angles, resulting in large differences in landform shapes, as shown in Fig. 9(a). Fig. 8(f) shows obvious changes in latitude, and all images exhibit significant distortion, which is consistent with the content of this study.

From Table II, it can be seen that there is a significant difference in pixel mean values between image pairs of site 3 and site 5, indicating a large disparity in overall brightness levels between these two sites. In addition, the pixel variance and standard deviation of image pairs from site 4 and site 5 are large, suggesting rich grayscale levels and distinct land cover types within these two sites.

In this article, three different algorithms are used to compare with the method proposed in this article to verify its advantages over the other algorithms, including SIFT [15], SAR-SIFT [21], and PSO-SIFT [47]. The feature-matching results of the four algorithms and the results of the registered checkboard overlay are shown. The comparison methods are briefly described as follows.

1) SIFT constructs a series of image pyramids by Gaussian filtering and downsampling, creating a multiscale space. The difference of Gaussian (DoG) operator is then used to detect extreme points in the scale space and a neighborhood window centered on the keypoints is used to extract local features within that window.
2) SAR-SIFT introduces the SAR-Harris operator to replace the DoG operator for keypoint detection. It uses circular descriptors to capture spatial context information.
3) PSO-SIFT is an image feature extraction algorithm that optimizes keypoint matching by combining particle swarm optimization (PSO) and SIFT.

The comparison experiments are conducted on the same hardware platform with Intel[1] Core[2] i5-9500 processor, where SIFT, SAR-SIFT, PSO-SIFT, and the algorithm in this article are run on MATLAB 2021a platform. The parameters of the SIFT, SAR-SIFT, and PSO-SIFT algorithms are kept at their optimal settings as described in their papers. All the above methods are experimented with the same image pairs as the method proposed in this article, and we use an improved RANSAC algorithm to remove the mismatching points.

B. Results of Comparative Experiments and Analyses

In order to evaluate the registration performance of the method in this article, we give the feature-matching results of different methods, as shown in Figs. 10–15. The quantitative comparison results are given in Table III.

From Fig. 10(a) and (b), it can be seen that SIFT and SAR-SIFT do not produce matching pairs in mountainous regions with large geometric distortion. PSO-SIFT matches fewer keypoints and obtains some incorrectly matching pairs labeled with blue boxes, as shown in Fig. 10(c). By visual inspection in Fig. 10(d), it can be observed that the feature-matching results obtained by the proposed method are quite accurate.

From Fig. 11(a) and (c), it can be seen that SIFT and PSO-SIFT obtain a small number of matching point pairs, some of which are incorrect. SAR-SIFT still produces few matching point pairs in mountainous regions with large geometric distortion, as shown in Fig. 11(b). Compared with other methods, the method in this article still obtains more matching point pairs, as shown in Fig. 11(d).

Fig. 12 shows the matching point pairs of different methods on site 3. As shown in Fig. 12(a) and (c), the matching point pairs of SIFT and PSO-SIFT are fewer in number and have false matches. Fig. 12(b) shows that SAR-SIFT produces matching point pairs in water where the features are not obvious. As shown in Fig. 12(d), the matching pairs obtained by this method are still uniform, abundant, and accurate.

From Fig. 13(a), (b), and (d), it can be seen that there are errors in the matches obtained by the SIFT. Only the SAR-SIFT and the method proposed in this article

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[2]Trademarked.

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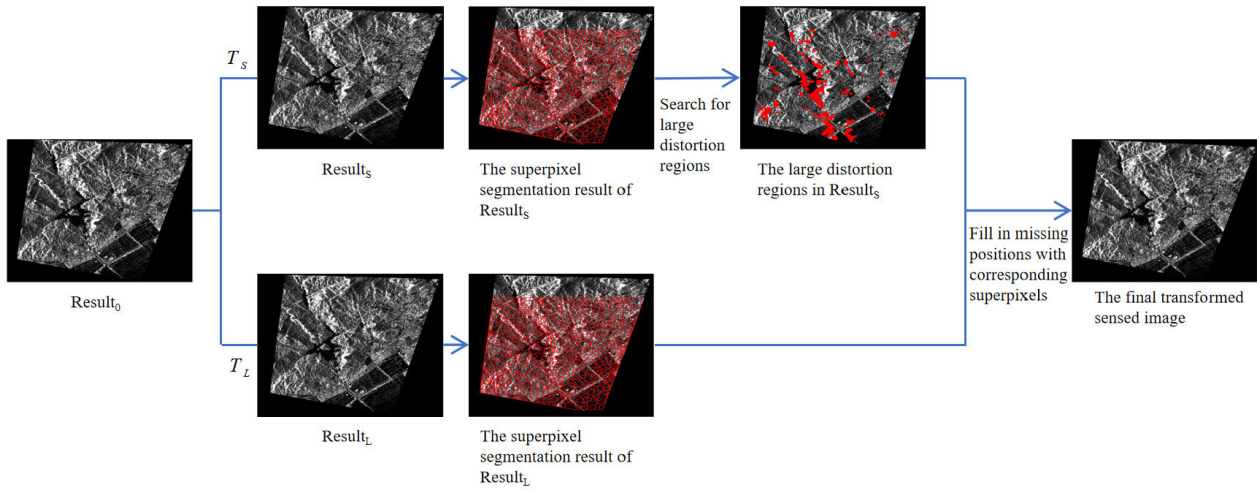


Fig. 7. Flowchart of superpixel replacement.

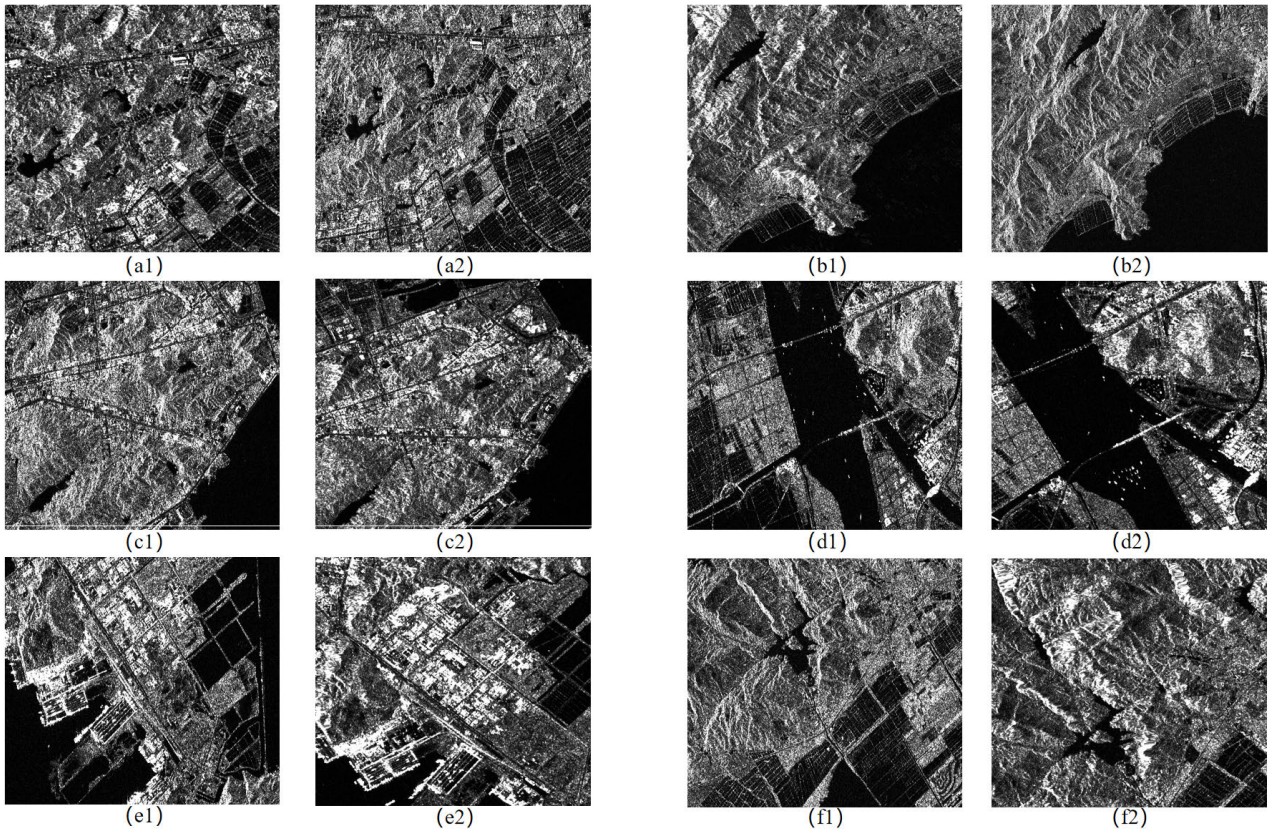


Fig. 8. Experimental SAR data of this paper. (a1)–(f1) Reference images of six sites. (a2)–(f2) Sensed images of six sites.

are able to obtain uniform and correct matching pairs, but there are still erroneous matches in the results of SAR-SIFT. However, as shown in Fig. 13(c), while the number of matching point pairs obtained by PSO-SIFT is still low.

In the registration results of site 5, SIFT produces concentrated and incorrect pairs as shown in Fig. 14(a). PSO-SIFT obtains the worst registration results as shown in Fig. 14(c). SAR-SIFT produces incorrectly matching pairs of points in the registration results as shown in Fig. 14(b). And the method still consistently obtains the most matching point pairs, as shown in Fig. 14(d).

We can see that PSO-SIFT obtains only a very small number of incorrectly matching point pairs for site 6 as shown in Fig. 15(c). Obviously, the rugged terrain causes great difficulties in the construction of PSO-SIFT feature descriptors. As shown by the blue boxes in Fig. 15(a) and (b), both SIFT and SAR-SIFT produce wrong matching point pairs. Yet our method still yields uniform and rich matching point pairs, as shown in Fig. 15(d).

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TABLE I
INFORMATION ABOUT THE SAR IMAGE PAIRS USED IN THE EXPERIMENT

Sensor	Site	Size	Resolution(Range×Azimuth)	Access time
Sentinel-1	Site 1	500 × 500(left) 500 × 500(right)	14m × 14m	20201010(left) 20211222(right)
	Site 2	700 × 700(left) 700 × 700(right)		
	Site 3	500 × 500(left) 600 × 500(right)		
	Site 4	500 × 500(left) 500 × 500(right)		
	Site 5	450 × 450(left) 480 × 480(right)		
	Site 6	500 × 500(left) 500 × 500(right)		

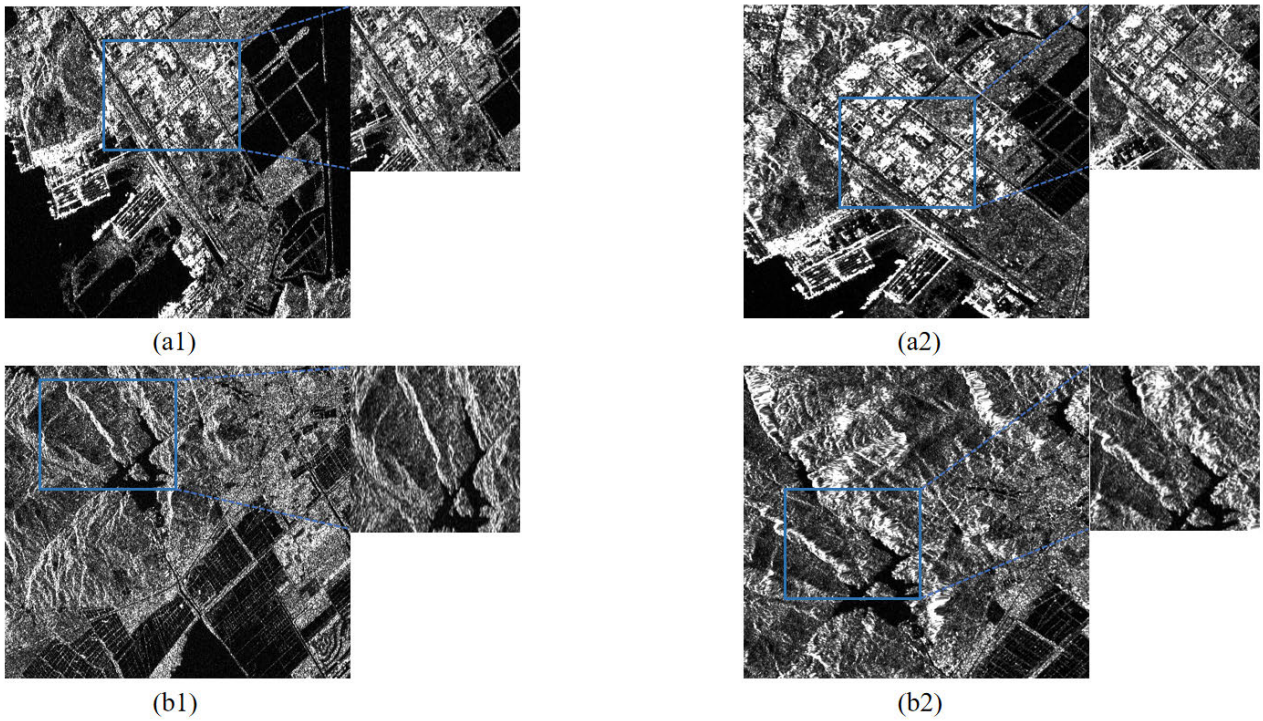


Fig. 9. Enlarged view of local details of some experimental SAR data. (a1) and (a2) Enlarged view of the reference image and the sensed image of Fig. 8(e). (b1) and (b2) Enlarged view of the reference image and the sensed image of Fig. 8(f).

TABLE II
PIXEL VALUE INFORMATION FOR IMAGE PAIRS (MEAN, STANDARD DEVIATION, AND VARIANCE)

	Mean Of Pixel Values	Std Of Pixel Values	Dev Of Pixel Values
Site1(left/right)	85.4/89.5	8.3/8.6	68.6/74.8
Site2(left/right)	67.5/72.8	8.1/8.4	66.3/70.3
Site3(left/right)	91.8/75.9	8.9/8.6	79.3/73.2
Site4(left/right)	64.4/60.7	8.6/8.5	73.4/73.0
Site5(left/right)	73.4/90.8	9.0/9.0	81.5/81.7
Site6(left/right)	87.7/82.3	8.5/8.2	71.5/67.9

Through the visual inspection in Figs. 10–15, SIFT cannot obtain accurate registration results. This is due to the fact that SAR image has speckle noise, which has a large impact on the feature extraction ability of SIFT. SAR-SIFT has stable registration performance on different SAR images. However, for SAR images with a large degree of distortion, its hand-crafted

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TABLE III

COMPARISON OF THE NCM POINT PAIRS, RMSE, AND RUNNING TIME (S) FOR SIX DIFFERENT METHODS OF SAR IMAGE PAIRS

method	Site1			Site2			Site3			Site4			Site5			Site6		
	NUM	RMSE	Time	NUM	RMSE	Time	NUM	RMSE	Time	NUM	RMSE	Time	NUM	RMSE	Time	NUM	RMSE	Time
SIFT	15	0.75	18.02	7	0.53	14.72	22	0.68	13.25	12	0.67	10.29	17	0.65	10.92	8	0.64	12.06
SAR-SIFT	17	0.63	24.49	13	0.47	48.74	26	0.61	31.11	17	0.61	25.22	20	0.54	20.62	17	0.59	22.49
PSO-SIFT	4	0.81	15.04	11	0.62	24.93	7	0.71	20.63	3	0.72	9.36	7	0.71	12.42	2	0.69	13.67
Our method	**28**	**0.61**	**19.63**	**27**	**0.45**	**27.33**	**26**	**0.58**	**19.5**	**23**	**0.57**	**14.32**	**27**	**0.51**	**16.0**	**23**	**0.55**	**18.44**

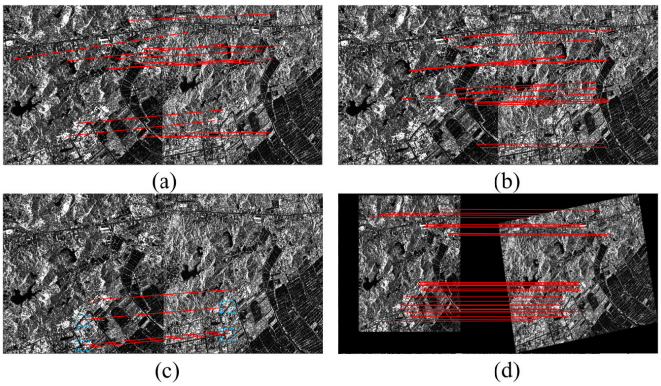


Fig. 10. Feature matching results of different methods for site 1. (a) SIFT. (b) SAR-SIFT. (c) PSO-SIFT. (d) Our proposed method.

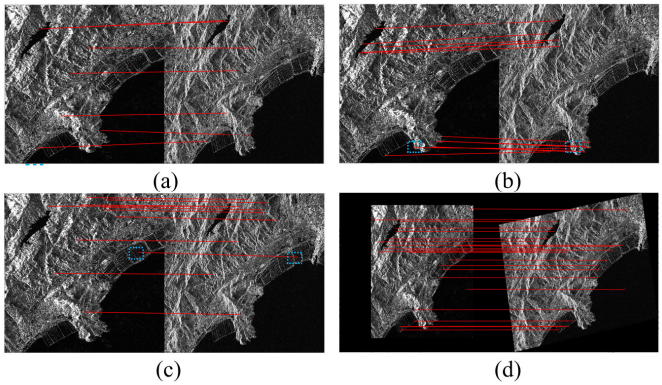


Fig. 11. Feature matching results of different methods for site 2. (a) SIFT. (b) SAR-SIFT. (c) PSO-SIFT. (d) Our proposed method.

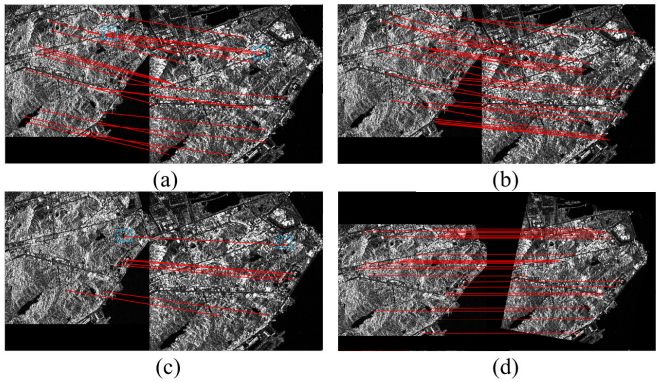


Fig. 12. Feature matching results of different methods for site 3. (a) SIFT. (b) SAR-SIFT. (c) PSO-SIFT. (d) Our proposed method.

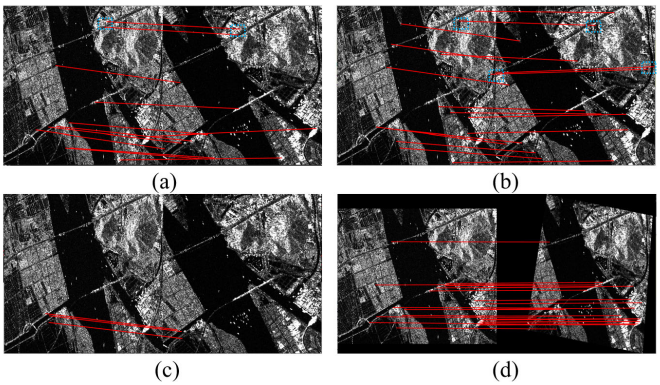


Fig. 13. Feature matching results of different methods for site 4. (a) SIFT. (b) SAR-SIFT. (c) PSO-SIFT. (d) Our proposed method.

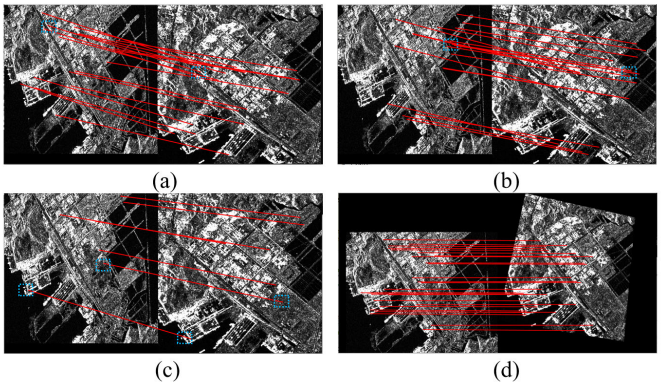


Fig. 14. Feature matching results of different methods for site 5. (a) SIFT. (b) SAR-SIFT. (c) PSO-SIFT. (d) Our proposed method.

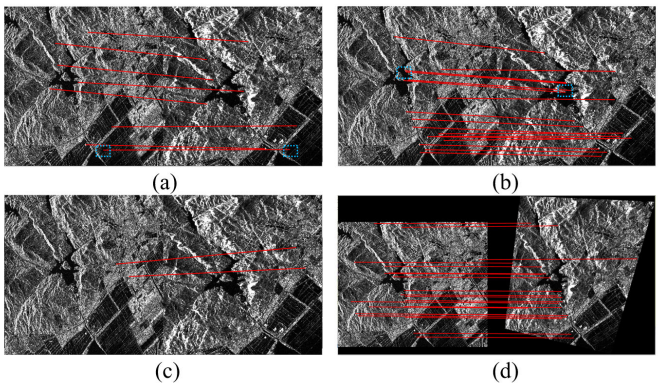


Fig. 15. Feature matching results of different methods for site 6. (a) SIFT. (b) SAR-SIFT. (c) PSO-SIFT. (d) Our proposed method.

feature descriptors still do not work well. It will result in fewer matching point pairs than the method in this article. Compared to other methods on the six sites, PSO-SIFT achieves the worst results. Obviously, PSO-SIFT has large limitations in its ability to be applied to different datasets. Compared with the other three methods, this article's method is able to obtain a larger number of more uniform and stable matching point pairs on different SAR images. It indicates that this article's method is able to obtain the keypoints with high repeatability and has a strong feature-matching capability.

Accordingly, we give the quantitative comparison results of different methods on six pairs of SAR images, as shown in Table III. We can get the same conclusion as the results obtained from visual inspection. PSO-SIFT achieves the worst

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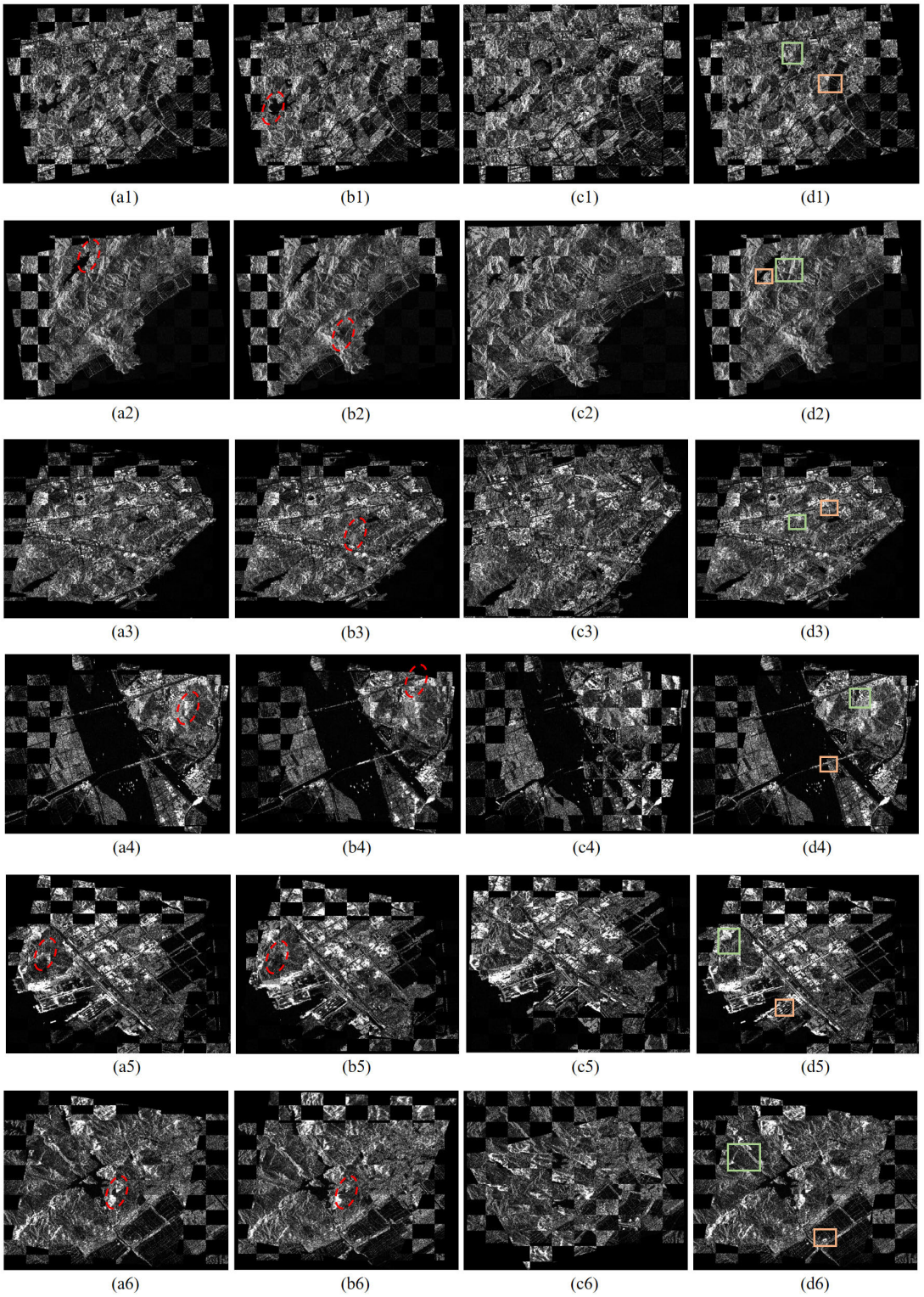


Fig. 16. Checkboard overlay diagram for different methods. (a1)–(a6) Registration results for SIFT. (b1)–(b6) Registration results for SAR-SIFT. (c1)–(c6) Registration results for PSO-SIFT. (d1)–(d6) Registration results for the proposed method. The green boxes indicate examples of large distortion regions, while the orange boxes represent examples of small distortion regions.

registration results on the six sites. The number of correctly matching (NCM) of this article's method is 11.5 times higher than that of PSO-SIFT on site 6 where it performs better. SIFT obtains good registration results on sites 3–5, while it performs poorly on sites 1, 2, and 6 where the terrains are more complex. The root mean square error (RMSE) of this article's method on site 5 is improved by 21.5% over the SIFT. The NCM of SIFT is also lower than that of SAR-SIFT and the method proposed in this article, indicating that its feature extraction capability cannot cope with complex sites with large distortions. SAR-SIFT can maintain a stable registration performance on six pairs of SAR images. Its average RMSE is only 6.3% worse than the method in this article. However, the running time of SAR-SIFT is much longer than that of other

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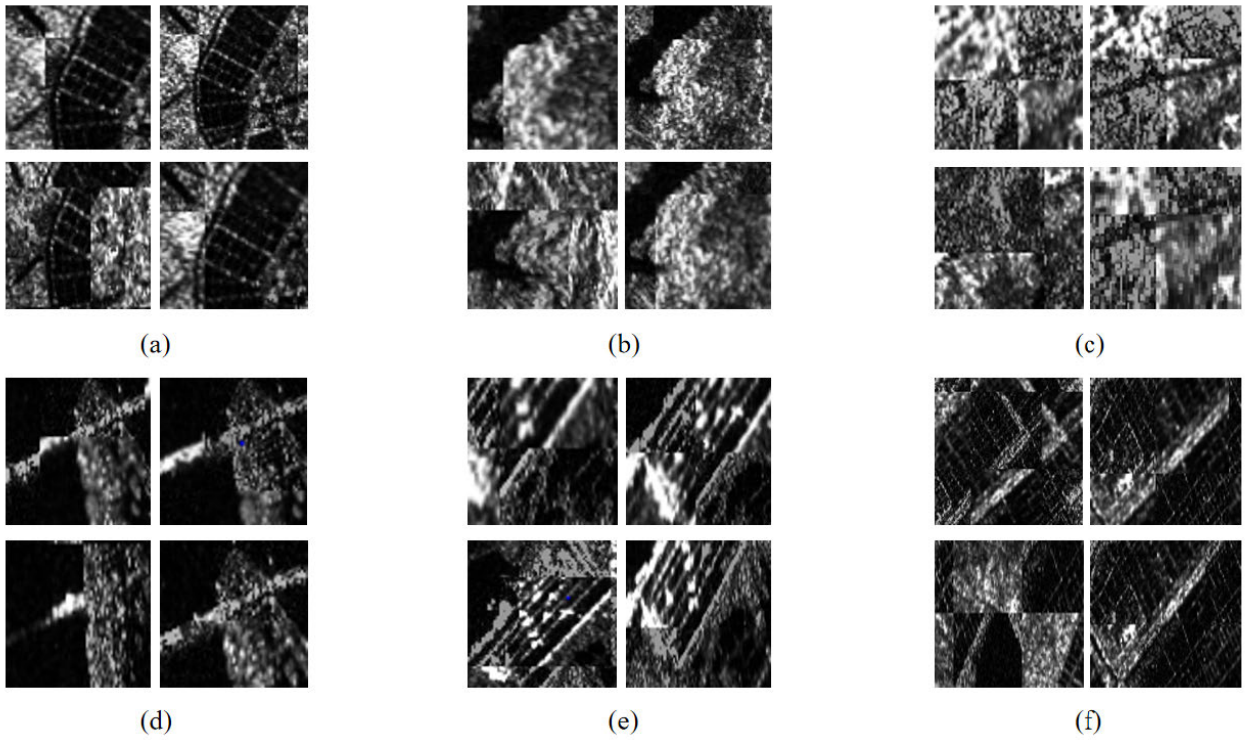


Fig. 17. Details of the small distortion regions. (a)–(f) Enlarged views of scenes within the orange boxes for six different sites, where the four detailed images for each scene, arranged from the top left to the bottom right, correspond to SIFT, SAR-SIFT, PSO-SIFT, and our proposed method, respectively.

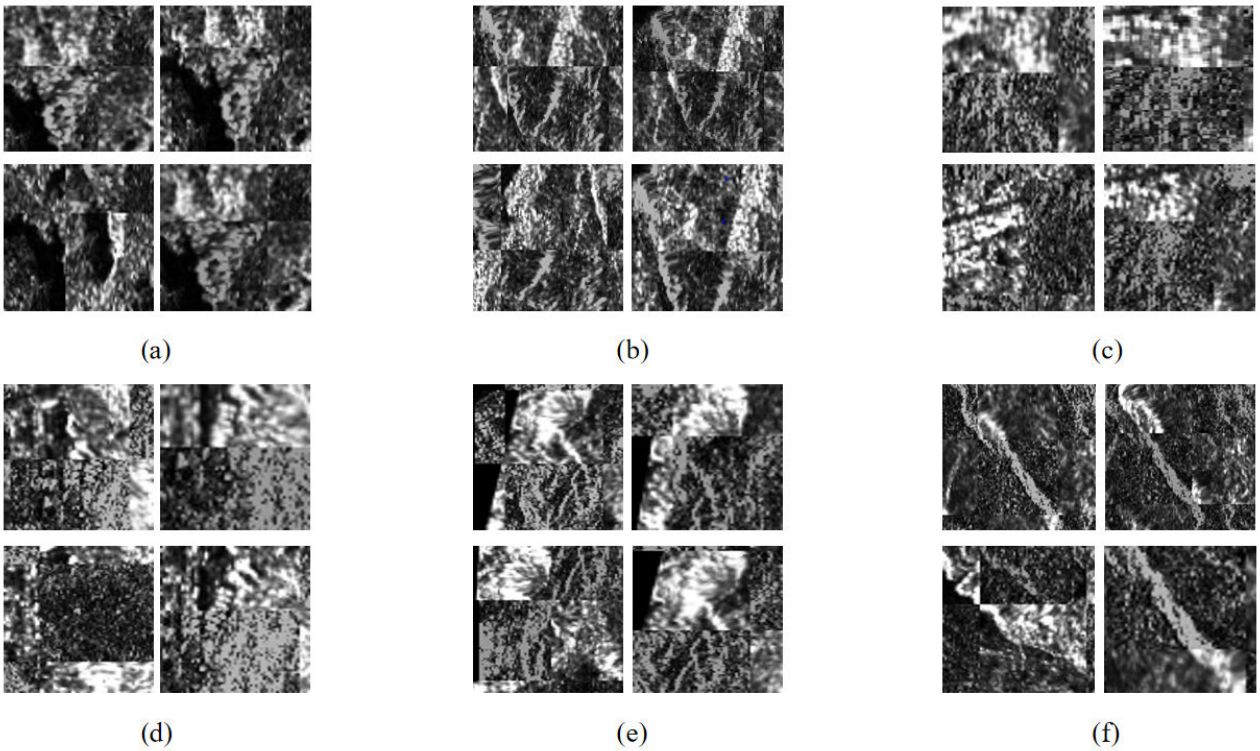


Fig. 18. Details of the large distortion regions. (a)–(f) Enlarged views of scenes within the green boxes for six different sites, where the four detailed images for each scene, arranged from the top left to the bottom right, correspond to SIFT, SAR-SIFT, PSO-SIFT, and our proposed method, respectively.

methods, which leads to the reduction of its application value. In the registration task of SAR images, both the number and accuracy of matching points have an important impact on the final registration results. By observing the registration results in terms of checkboard overlay in Fig. 16, we can see that when using the SAR-SIFT for registration, the estimation of the transform matrix is not accurate due to the small number of matching points, and some regions are still not fully registered.

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TABLE IV
NCM POINT PAIRS, RMSE, AND RUNNING TIME (S) OF SAR IMAGE PAIRS AFTER COARSE REGISTRATION

	NUM	RMSE	Time
Site1	8	0.77	21.45
Site2	9	0.60	25.66
Site3	15	0.70	18.42
Site4	6	0.70	12.68
Site5	12	0.69	14.78
Site6	5	0.65	15.64

Although the runtime of SAR-SIFT is longer compared to SIFT and PSO-SIFT, in contrast to the other two methods, SAR-SIFT exhibits a smaller RSME and a higher count of correctly matched point pairs. In addition, the number of points correctly matched by the SIFT, SAR-SIFT, and PSO-SIFT is less on-site 6 than on the other five sites. This is due to the fact that site 6 has large differences in terms of perspective and object shape relative to the other five sites, thus increasing the difficulty of feature matching. However, the method in this article still shows superior results relative to the other methods in this large distortion site. Our proposed method is more excellent than SAR-SIFT in terms of running time and superior to other advanced methods in terms of the NCM point pairs and RSME.

In order to show the registration results more intuitively, we present the registration results of SIFT, SAR-SIFT, and PSO-SIFT, as well as the method proposed in this article in the form of checkboard overlay, and outline with red dashed boxes some areas that are not registered, as shown in Fig. 16. We can observe that PSO-SIFT has the worst registration results on six sites. Due to the small number of matching point pairs, it is impossible to estimate an accurate transformation model, resulting in some regions not being fully registered. In Fig. 16(a1)–(a6), the checkboard results show more incoherent regions because SIFT has a small number of keypoint matching errors. As can be seen in Fig. 16(b1) and (b2), the checkboard results of SAR-SIFT show poor connectivity in the edges of the geographic features because its RMSE is larger than that of our proposed method. Our method outperforms the other comparative methods in terms of the number of matching point pairs and RSME. Thus, the checkboard results of the method in this article do not have significant errors.

Figs. 17 and 18 show, through local magnification, the details of the registration performance of SIFT, SAR-SIFT, PSO-SIFT, and our proposed method in regions with different distortion degrees. The continuity of farmland, roads, and coastlines can be observed in Fig. 17, while the continuity of mountain ranges is noticeable in Fig. 18. From the figures, it can be observed that the registration results of SIFT and PSO-SIFT are not satisfactory in both categories of distortion regions, likely due to the influence of SAR image noise on registration accuracy. As shown in Figs. 17(e) and 18(f), the registration results of SAR-SIFT exhibit poor connectivity in some small and large distortion regions, indicating unstable performance in registering SAR images with large distortion. Moreover, our proposed method achieves excellent matching results not only in regions with small geometric distortion but also demonstrates high registration accuracy in degrees with large geometric distortion by separately registering regions with different distortion degrees.

IV. DISCUSSION

From Tables IV and III, it can be observed that the results of NUM, RMSE, and time for image pairs after coarse registration alone are far from satisfactory, with only marginal improvement over the worst-performing PSO-SIFT method. This is because coarse registration can only establish a rough correspondence between images, and the results obtained at this stage are insufficient to be considered final. However, the fine registration method based on superpixels effectively enhances the overall performance of our proposed method, yielding results superior to other methods and achieving precise registration of SAR images with significant distortions.

V. CONCLUSION

This article proposes a large distortion SAR image registration method combined with superpixel segmentation. First, two SAR images are coarsely registered by geographic coordinate reference. Then, superpixel segmentation is performed on them after coarse registration. We obtain corresponding superpixels through the geometry mapping relationship of coarse registration, and their gray-scale similarity is calculated. Next, we use an adaptive threshold detector designed in this article, and each image is divided into two categories, regions with large and small geometric distortion degrees. Subsequently, we compute the transformation parameters based on matching feature points from different distortion regions. Registration is then performed separately on these two types of regions. Finally, the replacement of superpixel is performed to merge the two registration results, and we obtain the final registration result. The experimental results show that the method proposed in this article has higher registration accuracy compared to other traditional methods. It is also faster and more concise compared to deep learning methods. In the future, we will study the applicability of this method for processing other types of images such as optical and infrared. We will also make improvements to facilitate the application of multimodal remote sensing images.

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